IRIS Flower Classification using Logistic Regression and Random Forest

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1. Abstract

Automated taxonomic classification is an essential problem domain in modern computational biology. This study builds a highly reliable and reproducible workflow for plant species identification using advanced machine learning techniques. The famous Iris dataset, containing 150 total biological records across three distinct species, was utilized for empirical evaluation. Four key morphometric dimensions—sepal length, sepal width, petal length, and petal width—served as the continuous system features. The raw data was parsed into structured formats via custom Pandas Data Frame environments to extract statistical baselines.

The processed dataset was divided into an 80% training slice and a 20% testing subset using deterministic data splitting methods. Two distinct classification architectures—Logistic Regression (a parametric linear system) and Random Forest (a non-parametric tree-based ensemble)—were deployed to perform multi-class predictive modelling. Both learning frameworks were optimized and trained via Scikit-Learn tools. Performance was rigorously evaluated using multi-class precision, recall, F1-scores, and exact accuracy metrics.

Both machine learning frameworks successfully achieved a flawless 100% classification accuracy score on the test subset. The results validate that distinct structural properties provide strong predictive signals that allow clean mathematical differentiation. Ultimately, this implementation demonstrates high utility for automated botanical classification frameworks and establishes a clear benchmark for deployment in real-world environmental monitoring systems.

2. Introduction

2.1 Relevance of the Project

Manual plant species classification requires deep domain expertise and significant human time, creating a bottleneck in large-scale ecological studies. In modern environmental conservation, quick identification of flora is absolutely critical for biodiversity monitoring, habitat tracking, and ecosystem health assessments. Digital automation bridges this gap by leveraging automated data recognition techniques that eliminate human subjectivity. Computer systems can process multi-dimensional plant metrics drastically faster than human taxonomists, handling thousands of samples concurrently. This project establishes a robust baseline for comparing linear and non-linear classification strategies on biological data. The insights gained help in deploying scalable tools for environmental research applications and automated digital field guides.

2.2 Technology and Libraries Involved

- **Python Programming Language:** The underlying language used to execute data logic, model math, and training loops.
- **Pandas Library:** Used for high-performance tabular operations, multi-dimensional array slicing, and dataset restructuring.
- **Scikit-Learn (Sklearn):** The primary machine learning framework used for automated validation splits, evaluation metrics extraction, and model training.

- **Ensemble Tree Frameworks:** Implemented via Scikit-Learn's Random Forest module to handle non-linear decision boundaries through bagging mechanics.
- **Linear Estimators:** Implemented via multi-class Logistic Regression models to compute parametric optimization borders.

2.3 Background Material Survey and Literature Review

The Iris dataset was originally introduced by the famous British statistician and biologist Sir Ronald Fisher in his 1936 paper, "The use of multiple measurements in taxonomic problems." It is widely considered a gold-standard benchmark for testing classification and pattern recognition algorithms. The dataset contains precise measurements across three unique biological species variants:

1. *Iris Setosa*
2. *Iris Versicolour*
3. *Iris Virginica*

Historical literature highlights that one specific species (*Iris Setosa*) is entirely linearly separable from the other two due to its uniquely small petal metrics. Conversely, *Iris Versicolour* and *Iris Virginica* exhibit overlapping physical traits in multi-dimensional space, making their boundary separation highly non-linear. Over the decades, this dataset has been utilized to evaluate early linear discriminant functions, unsupervised cluster algorithms, and modern deep neural network layers. Understanding how different classifiers handle these mixed-boundary distributions remains a core component of pattern recognition research.

2.4 Technical Training Received

During the initial two weeks of the summer internship at IDEAS Foundation, ISI Kolkata, rigorous training modules were completed:

- Mastering high-performance data structure handling using vectorized Pandas operations and custom series filtering.
- Deep dive into descriptive statistical math, variance calculations, data distributions, and feature correlation matrix processing.
- Analytical theory regarding objective selection for data split validations to actively prevent model overfitting and data leakage.
- Supervised learning theory covering optimization algorithms, multi-class cross-entropy loss minimization, and ensemble decision tree theory.

3. Project Objective

The primary structural objectives established for this analytical study include:

- **Pipeline Automation:** Building an end-to-end reproducible machine learning data workflow from raw data ingestion to evaluation outputs.
- **Feature Engineering Evaluation:** Testing how raw continuous biological measurements translate to distinct class signals across different plant families.
- **Linear vs. Non-Linear Assessment:** Validating how generalized linear classifiers perform relative to tree-based ensemble models on overlapping data boundaries.
- **Statistical Performance Analysis:** Measuring multi-class validation boundaries using precision, recall, and exact F1 scores under rigorous testing constraints.
- **Overfitting Prevention:** Applying clean data split practices to confirm real-world generalization and model robustness.

4. Methodology

4.1 Data Pipeline Engineering

The structural mechanics of the analytical pipeline followed a clean data engineering architecture, moving systematically through raw extraction, conversion, feature isolation, training, and evaluation. The raw Iris dataset was pulled directly via the Scikit-Learn native dataset repository, removing the need for manual external CSV file fetching.

Once imported, the raw arrays were structured instantly into a unified Pandas DataFrame. The system mapped the continuous physical properties to four dedicated columns, while appending an integer-mapped target column at the index terminal. Feature-target isolation separated the structural metrics matrix from the class label vector cleanly. An 80/20 train-test split was executed to separate 120 samples for training optimization and 30 samples for unbiased validation testing. Finally, both models were fitted on the training split, and their weights were evaluated against the unseen test data.

4.2 Data Collection and Sampling Strategy

The data contains exactly 150 instances evenly distributed across three groups. Each species group contains exactly 50 target observations. The sampling methodology features equal, perfectly balanced representations to prevent prediction bias or minority class neglect during gradient updates.

Four physical attributes were recorded for each sample:

1. Sepal Length (measured in centimeters)
2. Sepal Width (measured in centimeters)
3. Petal Length (measured in centimeters)
4. Petal Width (measured in centimeters)

4.3 Data Preprocessing and Cleaning Steps

The dataset contains zero missing attribute values, removing the need for median or mean imputation steps. The text feature names were systematically transformed into structured lowercase string fields. Target records were converted to integer index labels to optimize processing efficiency:

- Class label `0` represents *Iris Setosa*
- Class label `1` represents *Iris Versicolour*
- Class label `2` represents *Iris Virginica*

4.4 Model Selection and Mathematical Logic

4.4.1 Logistic Regression Architecture

Logistic Regression calculates classification metrics by applying a multinomial Softmax function to the linear layer outputs. The algorithm maps continuous physical traits into a probability distribution bounded strictly between 0 and 1. The model parameters were optimized over a maximum budget of 200 iterations using the 'lbfgs' optimization solver to guarantee convergence across multi-class loss boundaries without early stopping errors.

4.4.2 Random Forest Classifier Architecture

The Random Forest model uses an ensemble of multiple independent Decision Trees to construct a robust non-parametric classifier. The algorithm creates 100 distinct decision tree estimators using bootstrap sampling techniques on the training subset. Predictions are determined by a majority vote across the ensemble trees. This approach minimizes individual tree variance and prevents overfitting on small datasets.

4.5 Validation Strategy

The feature matrix \(X\) and target vector \(y\) were separated cleanly. The data was divided into training and testing portions using an 80% to 20% ratio. The training step used 120 rows, while the validation step used 30 rows. Setting the random state seed to `42` ensures reproducible data splits across multiple execution environments.

5. Data Analysis And Result

5.1 Descriptive Analysis and Initial Summary

A summary review of the metric properties reveals distinct variations across features:

Parameter Field	Minimum Value	Maximum Value	Arithmetic Mean	Standard Deviation	Class Correlation
Sepal Length (cm)	4.3	7.9	5.84	0.83	0.7826
Sepal Width (cm)	2.0	4.4	3.05	0.43	-0.4194
Petal Length (cm)	1.0	6.9	3.76	1.76	0.9490 (High)
Petal Width (cm)	0.1	2.5	1.20	0.76	0.9565 (High)

The target column distribution confirmed an identical balanced spread:

- Class 0 (Iris Setosa): 50 records
- Class 1 (Iris Versicolour): 50 records
- Class 2 (Iris Virginica): 50 records

5.2 Model Performance and Metrics Comparison

5.2.1 Logistic Regression Performance

The trained linear model achieved an accuracy score of **1.00** (100% accuracy) on the testing slice. The optimization model achieved stable convergence across all classes without mis classifications, proving that the soft boundaries between the classes can be handled effectively by linear solvers when optimization iterations are sufficient.

5.2.2 Random Forest Classifier Performance

The ensemble tree system also reached a perfect performance score of **1.00**. Below is the detailed classification performance matrix extracted via the Scikit-Learn metrics engine:

Species Label Class	Precision Score	Recall Score	F1-Score Result	Validation Count	Support
0 (Iris-Setosa)	1.00	1.00	1.00	10	
1 (Iris-Versicolour)	1.00	1.00	1.00	9	
2 (Iris-Virginica)	1.00	1.00	1.00	11	
Macro Average Summary	1.00	1.00	1.00	30	
Weighted Average Summary	1.00	1.00	1.00	30	

5.3 Comparative Technical Breakdown

The model validation confirmed excellent performance across both architectures. This high accuracy is directly linked to the strong correlation scores of petal length (0.9490) and petal width (0.9565) relative to the target species. These clear feature distributions allow both linear separation planes and tree-based decision nodes to easily segment the test samples without showing signs of structural over fitting.

6. Conclusion

6.1 Core Findings and Justification

This project successfully deployed an automated machine learning workflow for biological classification tasks. Both models achieved perfect accuracy scores during testing operations. These results are supported by historical studies showing distinct physical differences between the three species. Petal properties serve as strong predictive indicators for classification, allowing basic algorithmic layers to achieve maximum performance.

6.2 Recommendations for Future Work

- Test the pipeline's stability using larger, web-scraped botanical datasets with high structural noise.
- Introduce synthetic noise into the feature metrics to assess model robustness under non-ideal data conditions.
- Evaluate model inference speeds across low-power edge computing devices to check usability for real-time mobile classification apps.

7. Appendices

Appendix 1: References

- Fisher, R. A. (1936). "The use of multiple measurements in taxonomic problems." *Annual Eugenics*, 7, Part II, 179-188.
- Duda, R. O., & Hart, P. E. (1973). *Pattern Classification and Scene Analysis*. John Wiley & Sons.
- Scikit-Learn Open-Source API documentation references for `linear_model` and `ensemble` modules.

Appendix 2: Digital Asset Repositories

- **GitHub Repository Link (Public):** https://github.com/anuran2003/-IRIS-Flower-Classification-using-Logistic-Regression-and-Random-Forest.git
- **Notebook GitHub Link (Direct File):** https://github.com/anuran2003/-IRIS-Flower-Classification-using-Logistic-Regression-and-Random-Forest/blob/main/01_IRIS_Flower_Classification_using_Logistic_Regression_and_Random_Forest_Summer_2026.ipynb